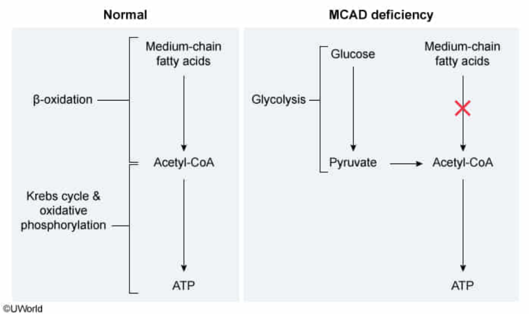
Medium-chain acyl-CoA dehydrogenase (MCAD) is an enzyme involved in β-oxidation and is encoded by the autosomal *ACADM* gene. MCAD deficiency is caused by a mutation in *ACADM* and results in impaired ability to digest medium-chain fatty acids for energy and in their subsequent accumulation. Which of the following interventions would best relieve the negative effects of MCAD deficiency?

- A. Mitochondrial gene therapy [17%]
- B. Supplementation with long-chain fatty acids [10%]
- C. Prolonged periods of fasting [8%]
- ✔ D. A high-carbohydrate, low-fat diet [63%]

Correct
63% answered correctly

Explanation:



Inborn errors of metabolism are genetic disorders caused by deleterious **mutations** in genes that code for **metabolic enzymes**. Because enzyme activity is decreased, these disorders result in the accumulation of metabolites upstream of the affected enzyme and decreased levels of the downstream products in the pathway. The resulting changes in metabolite levels frequently have negative clinical effects such as lethargy, impaired development, and toxicity.

According to the question, medium-chain acyl-CoA dehydrogenase (MCAD) is an enzyme involved in β-oxidation, a metabolic process that hydrolyzes fatty acids and eventually leads to ATP production. β-Oxidation is an important source of ATP during periods of fasting, after glycogen stores are exhausted. Because MCAD deficiency results in impaired β-oxidation, **therapies** should be aimed at restoring energy production, either by supplementing the deficient enzyme with a functional version or providing an **alternate fuel source** that follows a different metabolic pathway.

A **high-carbohydrate, low-fat diet** is an effective therapy for MCAD deficiency because carbohydrate breakdown similarly leads to ATP production. High-carbohydrate diets minimize the need for β-oxidation to break down either dietary or stored fatty acids. Furthermore, when fat intake is minimized, the residual activity from the impaired MCAD enzyme can decrease the levels of the accumulated metabolic intermediates.

(Choice A) Although β-oxidation occurs in the mitochondria, the gene that encodes *MCAD* (*ACADM*) is autosomal and therefore is encoded by nuclear DNA. Mitochondrial gene therapy targets the mitochondrial genome only; the mitochondrial genome lacks the regulatory elements to properly express MCAD, whereas the nuclear *ACADM* gene will not be repaired by mitochondrial therapy.

(Choice B) Long-chain fatty acids are digested into medium-chain fatty acids, which must be processed by MCAD. Supplementation with long-chain fatty acids in an MCAD-deficient patient would lead to an *increased* buildup of medium-chain fatty acids and yield little ATP.

(Choice C) β-Oxidation is an important energy source during periods of fasting. MCAD deficiency inhibits β-oxidation, so fasting is harmful to MCAD-deficient individuals.

Educational objective:
Inborn errors of metabolism are genetic disorders that result in impaired metabolic pathways. These disorders are typically treated either by supplementing the deficient enzyme with a functional version or by providing an alternate source of the desired metabolic products. Alternate sources must follow a separate, independent metabolic pathway from the defective one.

Time spent:0 sec	**QID**:401520
Subject: Biochemistry	**System:** Metabolic Reactions